Exploring natural mortality assumptions for the sardine assessment

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A range of alternative fixed natural mortality rates are tested for the Cold Temperate Sardine off the west coast of South Africa, together with a range of alternative fixed differences in the 0-year-old natural mortality rate between Cold Temperate Sardine off the west coast and Warm Temperate Sardine off the South Coast. The impact of assuming natural mortality increases by a fixed amount off the 'less preferred' coast for each stock is also considered. These results can be used to select fixed values for the baseline Operating Models and robustness tests for the upcoming Management Strategy Evaluation.

Keywords: assessment, natural mortality, sardine, South Africa

Background

The assessment of South African sardine using data up to the end of 2024 considered thus far has assumed fixed time-invariant rates of natural mortality for the Cold Temperate sardine (CTS) distributed off the west coast in line with previous assessments (de Moor 2025a,b). This document now explores a range of alternative values to inform the baseline and robustness Operating Models for the Management Strategy Evaluation of a joint sardine-anchovy Operational Management Procedure. In addition, following the initial results presented by de Moor (2025d), it is suggested that two further parameters associated with natural mortality are fixed instead of estimated and a range of these values is also explored herein.

Methods

Juvenile and adult natural mortality for CTS distributed off the west coast are fixed at a range of values from 0.8year^{-1} to 1.4year^{-1}, with the constraint that $M_{w,y,1+}^{CTS} \leq M_{w,y,0}^{CTS}$. The difference between $M_{w,y,0}^{CTS}$ and that of Warm Temperate sardine (WTS) recruits off the south coast is given by M_{juv}^{diff}. This was previously estimated with $M_{juv}^{diff} \sim U(-0.5{,}0.5)$, but here a range of fixed values for M_{juv}^{diff} of -0.25, 0, 0.25 and 0.5 are explored. The increase in natural mortality of all ages of CTS distributed off the west coast to those off the south coast and of WTS distributed off the south coast to those off the west coast is given by M_{inc}^{WTS}. This allows for an increase in natural mortality off the coast which is not a stocks' hypothesised preferred habitat (van der Lingen *et al*. 2023). This was previously estimated with $M_{inc}^{j} \sim U(0{,}1)$, but here fixed values of 0.0, 0.1 and 0.2 are explored, where the natural mortality of each stock does not change between coasts when $M_{inc}^{WTS} = 0$. All alternatives are considered for the assessment which does not assume a stock recruitment relationship during conditioning (de Moor 2025a), while due to time constraints, a smaller set of alternatives are considered for the assessment which assumes a Beverton-Holt stock recruitment relationship with $\sigma_{r,j}^2 = 5$ (de Moor 2025c).

Results and Discussion

The results are given in Tables 1 and 2 for $M_{inc}^{j} = 0.0$ and the combinations producing the better fits from these tables are compared against $M_{inc}^{j} = 0.1$ and $M_{inc}^{j} = 0.2$ in Table 3.

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The M_{ad}^{diff} is estimated to be similar (and thus $M_{s,y,1^+}^{WTS}$ is similar) for each combination of fixed natural mortalities between the scenario that does not assume a stock recruitment relationship and that which assumes a Beverton Holt stock recruitment relationship. Note that with some of the runs assuming a Beverton Holt stock recruitment relationship, the same objective function could be obtained with different stock recruitment parameters, given $\sigma_{r,j}^2 = 5$. This may need further consideration once baseline and robustness tests are selected.

Although there are a few exceptions, the fit to the data and the objective function are often very slightly poorer as M_{inc}^{WTS} increases from 0 to 0.1 and 0.2. It is thus suggested that any baseline hypotheses assume $M_{inc}^{WTS} = 0.0$ with one (set of) robustness tests assuming $M_{inc}^{WTS} = 0.2$ or $M_{inc}^{WTS} = 0.5$.

The combinations of natural mortality that provide the best fits to the data include:

$\boldsymbol{M_{w,y,0}^{CTS} = 0.8, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.3, M_{s,y,1^+}^{WTS} = 1.24 - 1.29}$

$M_{w,y,0}^{CTS} = 1.0, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.5, M_{s,y,1^+}^{WTS} = 1.19 - 1.28$

$\boldsymbol{M_{w,y,0}^{CTS} = 1.0, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.25, M_{s,y,1^+}^{WTS} = 1.25}$

$M_{w,y,0}^{CTS} = 1.2, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.45, M_{s,y,1^+}^{WTS} = 1.22 - 1.29$

$\boldsymbol{M_{w,y,0}^{CTS} = 1.2, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.20, M_{s,y,1^+}^{WTS} = 1.20}$

$M_{w,y,0}^{CTS} = 1.4, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.40, M_{s,y,1^+}^{WTS} = 1.07 - 1.31$

$\boldsymbol{M_{w,y,0}^{CTS} = 1.4, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.15, M_{s,y,1^+}^{WTS} = 1.15}$

And also

$M_{w,y,0}^{CTS} = 0.8, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.05, M_{s,y,1^+}^{WTS} = 1.05$

$M_{w,y,0}^{CTS} = 1.4, M_{w,y,1^+}^{CTS} = 0.8, M_{s,y,0}^{WTS} = 1.15, M_{s,y,1^+}^{WTS} = 1.15$

None of these correspond directly with that which was assumed for the baseline model of the previous hypothesis, which was 1.0year^{-1} for both juvenile and 1$^+$ fish of both components. The four options highlighted in bold above might be the closest on average, allowing for differences between ages and/or stocks. In the two cases above where the juvenile natural mortality is the same for both stocks, the adult natural mortality is estimated to be >0.4year^{-1} higher for WTS off the south coast compared to CTS off the west coast.

Model runs to consider other key model uncertainties to inform selections for the baseline and robustness tests will follow in a separate document.

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Table 1. The contributions to the objective function from likelihood (equations A29 – A33 of de Moor 2025a) and prior components for the model **without a stock recruitment relationship** together with some key parameter estimates at the joint posterior mode. Values given in **bold** are fixed and not estimated. Values in grey are derived from fixed/estimated parameters.

	$M^{CTS}_{w,y,0}$	**0.8**	**1.0**	**1.2**	**1.4**	**1.0**	**1.2**	**1.4**	**1.2**	**1.4**	**1.4**
	$M^{CTS}_{w,y,1^+}$	**0.8**	**0.8**	**0.8**	**0.8**	**1.0**	**1.0**	**1.0**	**1.2**	**1.2**	**1.4**
	$M^{WTS}_{s,y,0}$	**1.3**	**1.5**	**1.7**	**1.9**	**1.5**	**1.7**	**1.9**	**1.7**	**1.9**	**1.9**
	$M^{WTS}_{s,y,1^+}$	1.29	1.28	1.23	1.40	1.30	1.24	1.40	1.25	1.40	1.40
	M^{diff}_{juv}	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**
	M^{diff}_{ad}	0.01	0.22	0.47	0.50*	0.20	0.46	0.50*	0.45	0.50*	0.50*
	M^{WTS}_{inc}	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Objective function		-855.4	-855.3	-854.8	-851.9	-851.8	-851.2	-848.5	-847.2	-844.4	-841.2
log likelihood	$-lnL$	-904.6	-904.6	-904.2	-902.4	-901.3	-900.8	-899.1	-897.4	-895.1	-892.8
	$-lnL^{Nov}$	41.9	42.0	42.2	43.1	42.3	42.7	43.6	43.9	44.3	46.1
	$-lnL^{rec}$	30.8	30.8	31.0	30.7	31.1	31.3	31.2	32.2	31.5	32.9
	$-lnL^{sur\ propl}$	-496.1	-496.5	-496.7	-497.0	-494.4	-494.8	-495.2	-494.2	-493.7	-496.4
	$-lnL^{com\ propl}$	-481.1	-481.0	-480.7	-479.3	-480.3	-480.0	-478.7	-479.3	-477.2	-475.5
log priors	$ln(k^S_{ac})$	-0.5	-0.6	-0.7	-0.2	-0.6	-0.7	-0.2	-0.7	-0.2	-0.1
	$move_{y,1}$	4.1	4.1	4.1	4.1	4.0	4.0	4.1	4.0	3.9	3.8
	$\bar{l}_{1,y}$	41.9	41.8	41.8	42.0	42.0	41.9	42.1	42.1	42.3	42.3
	ϑ_a	0.8	0.8	0.8	0.8	0.8	0.7	0.8	0.7	0.7	0.7
	S_{50}	2.7	2.9	3.2	3.5	3.0	3.3	3.7	3.8	3.8	4.6
	$t_{0,c}$	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.1
	k^S_{ac}	0.68	0.70	0.78	0.94	0.70	0.79	0.94	0.79	0.95	0.95
	$k_{w,r}$	0.29	0.28	0.30	0.31	0.26	0.28	0.29	0.27	0.29	0.28
	$k_{s,r}$	0.28	0.27	0.29	0.30	0.26	0.28	0.29	0.27	0.28	0.27
	$M^{CTS}_{w,y,0}$	**0.8**	**1.0**	**1.2**	**1.4**	**1.0**	**1.2**	**1.4**	**1.2**	**1.4**	**1.4**
	$M^{CTS}_{w,y,1^+}$	**0.8**	**0.8**	**0.8**	**0.8**	**1.0**	**1.0**	**1.0**	**1.2**	**1.2**	**1.4**
	$M^{WTS}_{s,y,0}$	**1.05**	**1.25**	**1.45**	**1.65**	**1.25**	**1.45**	**1.65**	**1.45**	**1.65**	**1.65**
	$M^{WTS}_{s,y,1^+}$	1.05	1.25	1.29	1.25	1.25	1.32	1.27	1.30	1.25	1.21
	M^{diff}_{juv}	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**
	M^{diff}_{ad}	0.00	0.00	0.16	0.40	0.00	0.13	0.38	0.15	0.40	0.44
	M^{WTS}_{inc}	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Objective function		-853.0	-855.0	-855.0	-854.5	-851.1	-851.4	-844.8	-847.1	-846.9	-843.5
log likelihood	$-lnL$	-901.2	-904.0	-904.3	-903.9	-900.1	-900.9	-892.0	-896.6	-896.8	-893.6
	$-lnL^{Nov}$	42.8	41.9	41.9	42.1	42.3	42.2	41.7	43.4	43.8	44.5
	$-lnL^{rec}$	30.6	30.7	30.8	31.0	31.0	31.2	30.8	31.7	32.4	33.4
	$-lnL^{sur\ propl}$	-493.7	-495.6	-496.0	-496.2	-492.8	-494.0	-484.7	-492.5	-493.5	-493.2
	$-lnL^{com\ propl}$	-481.0	-481.1	-481.0	-480.7	-480.6	-480.3	-479.8	-479.2	-479.4	-478.2
log priors	$ln(k^S_{ac})$	-0.6	-0.5	-0.5	-0.7	-0.5	-0.5	-0.7	-0.5	-0.7	-0.7
	$move_{y,1}$	3.9	4.1	4.1	4.1	3.8	4.0	4.0	3.8	3.9	3.7
	$\bar{l}_{1,y}$	41.8	41.9	41.9	41.9	42.0	42.0	41.9	42.2	42.1	42.4
	ϑ_a	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.7
	S_{50}	2.1	2.6	2.9	3.1	2.6	3.0	1.0	3.1	3.6	3.7
	$t_{0,c}$	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
	k^S_{ac}	0.72	0.69	0.69	0.76	0.67	0.69	0.78	0.69	0.76	0.76
	$k_{w,r}$	0.37	0.29	0.27	0.29	0.28	0.25	0.27	0.25	0.26	0.26
	$k_{s,r}$	0.37	0.29	0.26	0.28	0.27	0.25	0.27	0.25	0.26	0.26

Table 1. (continued)

	$M_{w,y,0}^{CTS}$	**0.8**	**1.0**	**1.2**	**1.4**	**1.0**	**1.2**	**1.4**	**1.2**	**1.4**	**1.4**
	$M_{w,y,1^+}^{CTS}$	**0.8**	**0.8**	**0.8**	**0.8**	**1.0**	**1.0**	**1.0**	**1.2**	**1.2**	**1.4**
	$M_{s,y,0}^{WTS}$	**0.8**	**1.0**	**1.2**	**1.4**	**1.0**	**1.2**	**1.4**	**1.2**	**1.4**	**1.4**
	$M_{s,y,1^+}^{WTS}$	0.80	1.00	1.20	1.31	1.00	1.20	1.33	1.20	1.31	1.26
	M_{juv}^{diff}	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
	M_{ad}^{diff}	0.00	0.00	0.00	0.09	0.00	0.00	0.07	0.00	0.09	0.14
	M_{inc}^{WTS}	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Objective function		-844.9	-851.5	-854.4	-854.6	-847.3	-850.4	-851.0	-846.1	-846.6	-843.2
log likelihood	$-lnL$	-893.3	-899.6	-903.2	-903.9	-895.6	-899.4	-900.5	-895.2	-896.1	-893.2
	$-lnL^{Nov}$	45.9	43.1	42.0	41.8	44.0	42.5	42.3	43.8	43.5	44.4
	$-lnL^{rec}$	29.0	30.6	30.7	30.8	31.2	31.3	31.4	32.0	31.9	33.1
	$-lnL^{sur\ propl}$	-489.7	-492.6	-494.8	-495.6	-491.5	-492.6	-493.6	-491.7	-492.2	-492.3
	$-lnL^{com\ propl}$	-478.5	-480.7	-481.0	-481.0	-479.4	-480.7	-480.5	-479.4	-479.3	-478.5
log priors	$ln(k_{ac}^S)$	-0.6	-0.6	-0.6	-0.5	-0.6	-0.5	-0.5	-0.5	-0.5	-0.5
	$move_{y,1}$	4.2	3.8	4.0	4.1	3.8	3.9	4.0	3.7	3.8	3.8
	$\bar{l}_{1,y}$	42.0	41.8	41.9	41.9	42.0	42.0	42.0	42.3	42.3	42.4
	ϑ_a	0.7	0.8	0.8	0.8	0.8	0.7	0.7	0.7	0.7	0.7
	S_{50}	1.8	2.0	2.4	2.8	2.0	2.6	2.9	2.7	3.0	3.3
	$t_{0,c}$	0.3	0.3	0.2	0.2	0.3	0.2	0.2	0.2	0.2	0.2
	k_{ac}^S	0.82	0.73	0.69	0.68	0.74	0.67	0.68	0.68	0.68	0.67
	$k_{w,r}$	0.54	0.39	0.30	0.26	0.37	0.28	0.25	0.27	0.24	0.24
	$k_{s,r}$	0.53	0.39	0.30	0.26	0.37	0.28	0.25	0.27	0.24	0.23
	$M_{w,y,0}^{CTS}$	**0.8**	**1.0**	**1.2**	**1.4**	**1.0**	**1.2**	**1.4**	**1.2**	**1.4**	**1.4**
	$M_{w,y,1^+}^{CTS}$	**0.8**	**0.8**	**0.8**	**0.8**	**1.0**	**1.0**	**1.0**	**1.2**	**1.2**	**1.4**
	$M_{s,y,0}^{WTS}$	**0.55**	**0.75**	**0.95**	**1.15**	**0.75**	**0.95**	**1.15**	**0.95**	**1.15**	1.15
	$M_{s,y,1^+}^{WTS}$	0.55	0.75	0.95	1.15	0.75	0.95	1.15	0.95	1.15	1.15
	M_{juv}^{diff}	**-0.25**	**-0.25**	**-0.25**	**-0.25**	**-0.25**	**-0.25**	**-0.25**	**-0.25**	**-0.25**	**-0.25**
	M_{ad}^{diff}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	M_{inc}^{WTS}	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Objective function		-833.1	-839.5	-849.2	-853.6	-833.5	-845.5	-849.7	-839.9	-845.3	-843.0
log likelihood	$-lnL$	-880.2	-886.4	-897.2	-902.2	-880.9	-893.8	-898.7	-888.9	-894.5	-893.0
	$-lnL^{Nov}$	43.3	41.7	44.1	42.3	45.6	44.8	42.9	45.6	44.4	44.8
	$-lnL^{rec}$	26.5	26.8	30.1	30.8	28.9	31.4	31.6	31.2	32.1	33.4
	$-lnL^{sur\ propl}$	-476.8	-479.1	-492.3	-494.4	-478.3	-491.1	-492.8	-487.8	-492.0	-492.6
	$-lnL^{com\ propl}$	-473.3	-475.8	-479.2	-480.9	-477.1	-478.9	-480.4	-477.9	-478.9	-478.7
log priors	$ln(k_{ac}^S)$	-0.6	-0.6	-0.7	-0.6	-0.7	-0.6	-0.5	-0.6	-0.5	-0.5
	$move_{y,1}$	4.0	3.9	3.7	3.9	4.5	3.8	4.0	4.1	3.8	4.3
	$\bar{l}_{1,y}$	42.7	42.5	42.0	41.9	42.2	42.1	42.0	42.0	42.3	42.3
	ϑ_a	0.5	0.5	0.8	0.8	0.6	0.7	0.7	0.8	0.7	0.7
	S_{50}	0.3	0.3	1.9	2.4	0.5	2.0	2.5	2.4	2.7	2.9
	$t_{0,c}$	0.2	0.2	0.3	0.2	0.3	0.3	0.2	0.3	0.2	0.3
	k_{ac}^S	0.72	0.72	0.78	0.70	0.75	0.75	0.68	0.74	0.69	0.68
	$k_{w,r}$	0.53	0.43	0.43	0.32	0.46	0.39	0.29	0.36	0.28	0.27
	$k_{s,r}$	0.52	0.42	0.42	0.31	0.46	0.38	0.29	0.36	0.28	0.27

Table 2. The contributions to the objective function from likelihood (equations A29 – A33 of de Moor 2025a) and prior components for the model **assuming a Beverton Holt stock recruitment relationship with $\sigma^2_{r,j} = 5$** together with some key parameter estimates at the joint posterior mode. Values given in **bold** are fixed and not estimated. Values in grey are derived from fixed/estimated parameters.

	$M^{CTS}_{w,y,0}$	**0.8**	**1.0**	**1.2**	**1.4**	**1.0**	**1.2**	**1.4**	**1.2**	**1.4**	**1.4**
	$M^{CTS}_{w,y,1^+}$	**0.8**	**0.8**	**0.8**	**0.8**	**1.0**	**1.0**	**1.0**	**1.2**	**1.2**	**1.4**
	$M^{WTS}_{s,y,0}$	**1.3**	**1.5**	**1.7**	**1.9**	**1.5**	**1.7**	**1.9**	**1.7**	**1.9**	**1.9**
	$M^{WTS}_{s,y,1^+}$	1.27	1.26	1.20	1.40	1.26	1.28	1.40	1.28	1.40	1.40
	M^{diff}_{juv}	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**	**0.50**
	M^{diff}_{ad}	0.03	0.24	0.5*	0.50*	0.24	0.42	0.50*	0.42	0.50*	0.50*
	M^{WTS}_{inc}	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Objective function		-778.0	-777.8	-777.6	-777.0	-774.2	-774.7*	-767.5*	-771.2*	-771.2	-767.9
log likelihood	$-lnL$	-901.8	-901.9	-901.9	-901.8	-897.9	-899.0	-890.4	-895.2	-896.0	-892.5
	$-lnL^{Nov}$	42.0	42.2	42.3	42.2	42.5	42.5	41.6	43.2	43.0	43.6
	$-lnL^{rec}$	31.1	31.1	31.1	31.4	31.0	31.5	31.1	31.9	32.1	32.8
	$-lnL^{sur\ propl}$	-495.2	-495.6	-495.9	-496.1	-492.9	-494.0	-484.7	-492.3	-492.8	-491.8
	$-lnL^{com\ propl}$	-479.6	-479.6	-479.5	-479.3	-478.5	-479.0	-478.4	-478.0	-478.3	-477.1
log priors	$ln(k^S_{ac})$	-0.7	-0.7	-0.7	-0.6	-0.7	-0.6	-0.6	-0.6	-0.5	-0.5
	$move_{y,1}$	4.0	4.0	4.0	4.0	3.8	4.0	4.0	3.8	4.0	3.9
	$\bar{l}_{1,y}$	42.2	42.2	42.2	42.3	42.4	42.3	42.3	42.4	42.5	42.6
	ϑ_a	0.8	0.8	0.8	0.8	0.8	0.7	0.8	0.7	0.7	0.7
	S_{50}	2.7	2.9	3.2	3.4	2.9	3.2	1.7	3.3	3.7	3.8
	ε^j_y	74.6	74.6	74.6	74.7	74.3	74.5	74.4	74.1	74.3	73.9
	$t_{0,c}$	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
	k^S_{ac}	0.74	0.74	0.74	0.73	0.74	0.72	0.70	0.71	0.69	0.69
	$k_{w,r}$	0.31	0.30	0.28	0.24	0.29	0.25	0.22	0.25	0.21	0.20
	$k_{s,r}$	0.31	0.29	0.28	0.23	0.29	0.25	0.21	0.24	0.20	0.20
	$M^{CTS}_{w,y,0}$	**0.8**	**1.0**	**1.2**	**1.4**	**1.0**	**1.2**	**1.4**	**1.2**	**1.4**	**1.4**
	$M^{CTS}_{w,y,1^+}$	**0.8**	**0.8**	**0.8**	**0.8**	**1.0**	**1.0**	**1.0**	**1.2**	**1.2**	**1.4**
	$M^{WTS}_{s,y,0}$	**1.05**	**1.25**	**1.45**	**1.65**	**1.25**	**1.45**	**1.65**	**1.45**	**1.65**	**1.65**
	$M^{WTS}_{s,y,1^+}$	1.05	1.25	1.27	1.27	1.25	1.27	1.30	1.29	1.29	1.24
	M^{diff}_{juv}	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**	**0.25**
	M^{diff}_{ad}	0.00	0.00	0.18	0.38	0.00	0.18	0.35	0.16	0.36	0.41
	M^{WTS}_{inc}	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**
Objective function		-775.3	-777.2*	-777.5	-777.3	-773.6	-773.7	-774.3*	-770.6	-771.2	-768.1*
log likelihood	$-lnL$	-898.7	-900.9	-901.6	-901.6	-896.9	-897.5	-898.6	-894.3	-895.7	-892.2
	$-lnL^{Nov}$	44.4	42.0	42.1	42.2	42.5	42.5	42.4	43.2	43.3	43.7
	$-lnL^{rec}$	29.1	31.0	31.1	31.1	31.0	31.1	31.6	32.1	32.7	33.6
	$-lnL^{sur\ propl}$	-492.7	-494.2	-495.2	-495.5	-491.9	-492.7	-493.7	-491.4	-493.1	-492.2
	$-lnL^{com\ propl}$	-479.5	-479.6	-479.6	-479.5	-478.7	-478.5	-479.0	-478.2	-478.5	-477.3
log priors	$ln(k^S_{ac})$	-0.7	-0.7	-0.6	-0.7	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
	$move_{y,1}$	3.9	4.0	4.0	4.0	3.7	3.8	4.0	3.7	4.1	3.8
	$\bar{l}_{1,y}$	42.1	42.2	42.3	42.3	42.4	42.4	42.3	42.6	42.4	42.6
	ϑ_a	0.8	0.7	0.8	0.8	0.8	0.8	0.7	0.7	0.7	0.7
	S_{50}	2.8	2.5	2.9	3.1	2.6	2.9	3.2	3.1	3.5	3.6
	ε^j_y	74.3	74.6	74.6	74.6	74.2	74.3	74.5	74.0	74.1	73.7
	$t_{0,c}$	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
	k^S_{ac}	0.77	0.74	0.74	0.74	0.73	0.73	0.71	0.70	0.70	0.70
	$k_{w,r}$	0.40	0.32	0.29	0.27	0.30	0.28	0.25	0.25	0.23	0.23
	$k_{s,r}$	0.40	0.31	0.29	0.27	0.30	0.28	0.25	0.25	0.23	0.23

Table 2. (continued)

	$M_{w,y,0}^{CTS}$	**0.8**	**1.0**	**1.2**	**1.4**						
	$M_{w,y,1^+}^{CTS}$	**0.8**	**0.8**	**0.8**	**0.8**						
	$M_{s,y,0}^{WTS}$	**0.8**	**1.0**	**1.2**	**1.4**						
	$M_{s,y,1^+}^{WTS}$	0.80	1.00	1.20	1.29						
	M_{juv}^{diff}	**0.00**	**0.00**	**0.00**	**0.00**						
	M_{ad}^{diff}	0.00	0.00	0.00	0.11						
	M_{inc}^{WTS}	**0.00**	**0.00**	**0.00**	**0.00**						
Objective function		-768.9	-774.2	-777.0*	-777.1*						
log likelihood	$-lnL$	-891.6	-896.7	-900.6	-901.2						
	$-lnL^{Nov}$	46.1	43.5	42.2	41.9						
	$-lnL^{rec}$	29.1	30.7	31.0	31.2						
	$-lnL^{sur\ propl}$	-489.5	-492.3	-494.1	-494.8						
	$-lnL^{com\ propl}$	-477.2	-478.7	-479.6	-479.5						
log priors	$ln(k_{ac}^{S})$	-0.6	-0.6	-0.7	-0.6						
	$move_{y,1}$	4.1	3.7	4.0	4.0						
	$\bar{l}_{1,y}$	42.3	42.2	42.3	42.3						
	ϑ_a	0.7	0.8	0.7	0.8						
	S_{50}	1.9	2.0	2.5	2.8						
	ε_y^j	73.8	74.2	74.6	74.6						
	$t_{0,c}$	0.3	0.3	0.2	0.2						
	k_{ac}^{S}	0.84	0.80	0.74	0.73						
	$k_{w,r}$	0.55	0.43	0.32	0.29						
	$k_{s,r}$	0.55	0.43	0.32	0.28						

Table 3. The objective function and negative log likelihood for a range of fixed values for $M^{CTS}_{w,y,0}$, $M^{CTS}_{w,y,1^+}$ and M^{diff}_{juv}, selected from Table 1, and showing the difference for values of 0.0, 0.1 and 0.2 for M^{WTS}_{inc}.

			No stock recruitment relationship						Beverton Holt stock recruitment relationship				
$M^{CTS}_{w,y,0}$	$M^{CTS}_{w,y,1^+}$	M^{diff}_{juv}	$M^{WTS}_{s,y,0}$	$M^{WTS}_{s,y,1^+}$	M^{diff}_{ad}	M^{WTS}_{inc}	Obj Fn	$-lnL$	$M^{WTS}_{s,y,0}$	$M^{WTS}_{s,y,1^+}$	M^{diff}_{ad}	Obj Fn	$-lnL$
0.8	**0.8**	**0.5**	1.3	1.29	0.01	**0.0**	-855.4	-904.6	1.3	1.27	0.03	-778.0	-901.8
				1.27	0.03	**0.1**	-855.2	-904.3		1.25	0.05	-777.7*	-901.5
				1.24	0.06	**0.2**	-854.9	-904.0		1.24	0.06	-777.4	-901.3
1.0	**0.8**		1.5	1.28	0.22	**0.0**	-855.3	-904.6	1.5	1.26	0.24	-777.8	-901.9
				1.24	0.26	**0.1**	-854.9	-904.1		1.24	0.26	-777.5*	-901.6
				1.19	0.31	**0.2**	-854.4	-903.6		1.23	0.27	-777.1	-901.2
1.0	**1.0**		1.5	1.30	0.20	**0.0**	-851.8	-901.3	1.5	1.26	0.24	-774.2	-897.9
				1.21	0.29	**0.1**	-850.9	-900.3		1.28	0.22	-774.5	-898.6
				1.21	0.29	**0.2**	-850.9	-900.3		1.26	0.24	-774.3*	-898.5
0.8	**0.8**	**0.25**	1.05	1.05	0.00	**0.0**	-853.0	-901.2	1.05	1.05	0.00	-775.3	-898.7
				1.05	0.00	**0.1**	-853.3	-901.6		1.05	0.00	-776.0*	-899.0
				1.05	0.00	**0.2**	-853.3	-901.7		1.05	0.00	-775.1*	-897.9
1.0	**0.8**		1.25	1.25	0.00	**0.0**	-855.0	-904.0	1.25	1.25	0.00	-777.2*	-900.9
				1.25	0.00	**0.1**	-854.8	-903.8		1.25	0.00	-777.2	-901.1
				1.25	0.00	**0.2**	-854.5	-903.6		1.25	0.00	-777.0*	-901.0
1.2	**0.8**		1.45	1.29	0.16	**0.0**	-855.0	-904.3	1.45	1.27	0.18	-777.5	-901.6
				1.26	0.19	**0.1**	-854.6	-903.9		1.26	0.19	-777.1	-901.2
				1.22	0.23	**0.2**	-854.2	-903.4		1.24	0.21	-776.9	-901.1
1.0	**1.0**		1.25	1.25	0.00	**0.0**	-851.1	-900.1	1.25	1.25	0.00	-773.6	-896.9
				1.25	0.00	**0.1**	-851.2	-900.4		1.24	0.01	-773.2*	-896.5
				1.25	0.00	**0.2**	-850.8	-900.0		1.22	0.03	-772.7	-896.0
1.2	**1.0**		1.45	1.32	0.13	**0.0**	-851.4	-900.9	1.45	1.27	0.18	-773.7	-897.5
				1.28	0.17	**0.1**	-851.1	-900.5		1.26	0.19	-773.4	-897.2
				1.24	0.21	**0.2**	-850.6	-900.0		1.28	0.17	-773.8	-898.0
1.2	**1.2**		1.45	1.30	0.15	**0.0**	-847.1	-896.6	1.45	1.29	0.16	-770.6	-894.3
				1.29	0.16	**0.1**	-846.7	-896.1		1.29	0.16	-770.3	-894.0
				1.25	0.20	**0.2**	-846.2	-895.6		1.28	0.17	-769.9	-893.7
0.8	**0.8**	**0.0**	0.80	0.80	0.00	**0.0**	-844.9	-893.3	0.80	0.80	0.00	-768.9	-891.6
				0.80	0.00	**0.1**	-844.1	-891.8		0.80	0.00	-766.7*	-889.5
				0.80	0.00	**0.2**	-841.3	-887.6		0.80	0.00	-767.5*	-889.0
1.0	**0.8**		1.00	1.00	0.00	**0.0**	-851.5	-899.6	1.00	1.00	0.00	-774.2	-896.7
				1.00	0.00	**0.1**	-851.6	-899.7		1.00	0.00	-774.0*	-896.6
				1.00	0.00	**0.2**	-851.7	-899.8		1.00	0.00	-773.1*	-896.2
1.2	**0.8**		1.20	1.20	0.00	**0.0**	-854.4	-903.2	1.20	1.20	0.00	-777.0*	-900.6
				1.20	0.00	**0.1**	-854.2	-903.0		1.14	0.06	-776.2	-899.6
				1.20	0.00	**0.2**	-854.0	-903.0		1.08	0.12	-777.4	-901.4
1.4	**0.8**		1.40	1.31	0.09	**0.0**	-854.6	-903.9	1.40	1.29	0.11	-777.1*	-901.2
				1.28	0.12	**0.1**	-854.3	-903.5		1.18	0.22	-777.2*	-901.4
				1.24	0.16	**0.2**	-853.9	-903.1		1.07	0.33	-777.3	-901.6
1.0	**1.0**		1.00	1.00	0.00	**0.0**	-847.3	-895.6	1.00				
				1.00	0.00	**0.1**	-850.9	-900.3					
				1.00	0.00	**0.2**	-847.2	-895.4					
1.2	**1.0**		1.20	1.20	0.00	**0.0**	-850.4	-899.4	1.20				
				1.20	0.00	**0.1**	-850.2	-899.2					
				1.20	0.00	**0.2**	-850.4	-899.4					
1.4	**1.0**		1.40	1.33	0.07	**0.0**	-851.0	-900.5	1.40				
				1.30	0.10	**0.1**	-850.7	-899.9					
				1.25	0.15	**0.2**	-850.2	-899.4					
1.2	**1.2**		1.20	1.20	0.00	**0.0**	-846.1	-895.2	1.20				
				1.20	0.00	**0.1**	-845.7	-894.8					
				1.20	0.00	**0.2**	-845.4	-894.5					
1.4	**1.2**		1.40	1.31	0.09	**0.0**	-846.6	-896.1	1.40				
				1.29	0.11	**0.1**	-846.1	-895.5					
				1.27	0.13	**0.2**	-845.6	-895.0					
1.4	**1.4**		1.40	1.26	0.14	**0.0**	-843.2	-893.2	1.40				
				1.25	0.15	**0.1**	-842.4	-892.3					
				1.26	0.14	**0.2**	-841.8	-891.7					

Table 3. (continued)

			No stock recruitment relationship						Beverton Holt stock recruitment relationship				
$M_{w,y,0}^{CTS}$	$M_{w,y,1^+}^{CTS}$	M_{juv}^{diff}	$M_{s,y,0}^{WTS}$	$M_{s,y,1^+}^{WTS}$	M_{ad}^{diff}	M_{inc}^{WTS}	Obj Fn	$-lnL$	$M_{s,y,0}^{WTS}$	$M_{s,y,1^+}^{WTS}$	M_{ad}^{diff}	Obj Fn	$-lnL$
1.2	**0.8**	**-0.25**	0.95	0.95	0.00	**0.0**	-849.2	-897.2					
				0.95	0.00	**0.1**	-849.1	-897.1					
				0.95	0.00	**0.2**	-849.2	-897.3					
1.4	**0.8**		1.15	1.15	0.00	**0.0**	-853.6	-902.2					
				1.15	0.00	**0.1**	-853.4	-902.1					
				1.15	0.00	**0.2**	-853.3	-902.0					
1.2	**1.0**		0.95	0.95	0.00	**0.0**	-845.5	-893.8					
				0.95	0.00	**0.1**	-845.2	-893.4					
				0.95	0.00	**0.2**	-845.1	-893.2					
1.4	**1.0**		1.15	1.15	0.00	**0.0**	-849.7	-898.7					
				1.15	0.00	**0.1**	-849.5	-898.5					
				1.15	0.00	**0.2**	-849.4	-898.4					
1.4	**1.2**		1.15	1.15	0.00	**0.0**	-845.3	-894.5					
				1.15	0.00	**0.1**	-844.5	-893.7					
				1.15	0.00	**0.2**	-844.3	-893.4					